

System Architecture & Service Level Agreement

1. Project Overview

This document details the architectural specifications, implemented features, and operational terms for the multi-location Inventory and Sales Management System. The application is a high-performance, cloud-native solution designed to streamline wholesale and retail operations across central warehouses and distributed shop locations.

2. Technology Stack & Infrastructure

The system is built on a modern, enterprise-grade technology stack ensuring scalability, security, and real-time performance.

- **Frontend Framework:** React.js with TypeScript (enabling strict type-safety and robust component architecture).
- **User Interface (UI):** Tailwind CSS, providing a responsive, clean, and highly customizable design language.
- **State Management:** Zustand, offering lightning-fast global state synchronization across all modules.
- **Backend Database:** Google Firebase Firestore, a highly scalable NoSQL cloud database providing real-time data synchronization and offline support.
- **Hosting & Edge Deployment:** Cloudflare Pages, delivering the application globally through edge servers for maximum speed and 99.99% uptime reliability.
- **Data Processing:** SheetJS (XLSX) library utilized for complex Excel import/export processing and local data parsing.

3. System Modules & Provided Features

- **Multi-Location Architecture:** Complete isolation and aggregation of data between Central Warehouses and distinct Shop locations.
- **Intelligent Inventory Engine:** Real-time tracking of Opening, Received, Supplied, Returned, and Closing balances for thousands of unique SKUs.
- **Global Import Utility:** Bulk Excel data ingestion with intelligent SKU deduplication, automated discrepancy handling, and real-time cost basis updating.
- **Sales & Point of Sale (POS):** Streamlined checkout interface supporting custom cart logic, dynamic discounting, and automated receipt generation.
- **Transfers & Returns:** Auditable, secure stock movement between locations, and robust handling of customer returns to active inventory.
- **Advanced Reporting:** Generation of Daily Sales Reports and comprehensive Global Stock Reports. Features include multi-format extraction (Clean PDF and Excel formats) with dynamic pagination.

4. Terms and Conditions

The following Service Level Agreement (SLA) outlines the conditions regarding ongoing maintenance and support for the deployed application:

- **Error Resolution (Bug Fixes):** Any functional errors, bugs, or system faults directly originating

from the agreed-upon initial deployment scope will be investigated and resolved at **no additional charge**.

- **Maintenance & Future Modifications:** Any requests for structural modifications, new feature additions, layout redesigns, or architectural changes made **after** the final deployment will be considered separate work orders. These will be **charged accordingly** based on the specific requirements, complexity, and estimated hours of the requested work.

By utilizing this system, all parties acknowledge and agree to the operational and maintenance terms outlined above.